SANGBAY
Gas Sensor

GT-TS01A

PRODUCTS MANUAL

1.Introduction

The GT-TSO1A laser methane sensor is developed based on Tunable Semiconductor Laser Absorption Spectrum (TDLAS) technology. It uses a 1653.7nm DFB laser to detect the characteristic "fingerprint spectrum" absorption of methane gas, resisting interference from water vapor or other gases. It adopts a compact multi-reflection optical path structure design, resulting in a long absorption optical path and high detection sensitivity and accuracy. It selects laser and photodetector devices with excellent high and low temperature performance and employs wide-temperature compensation and pressure compensation algorithms, making it highly adaptable to various environments. It adopts a standard threaded mounting interface and a standard communication interface design, making it highly versatile.

As the core component of a methane gas monitor, this product is used for methane gas leak detection and alarm. It can be widely used in methane gas leak detection and alarm applications in underground spaces, catering industry, industrial chemicals, underground valve wells, and coal mines.



2. Characteristics

★ Methane characteristic "fingerprint" absorption detection provides unique identification of methane gas, unaffected by interference from other gases or water vapor;
★ Non-destructive absorption detection, resistant to poisoning, and long lifespan;
★ Laser detection offers high stability and requires no calibration;
★ Long effective absorption optical path, achieving a long effective absorption optical path in a compact structure, resulting in high detection accuracy and sensitivity;
★ Wide bandwidth temperature compensation correction algorithm and pressure compensation algorithm, providing strong adaptability to high and low temperatures and different pressure environments;
★ Standard interface design ensures high versatility;
★ Intrinsically safe explosion-proof design and EMC protection design ensure stable operation under harsh conditions.

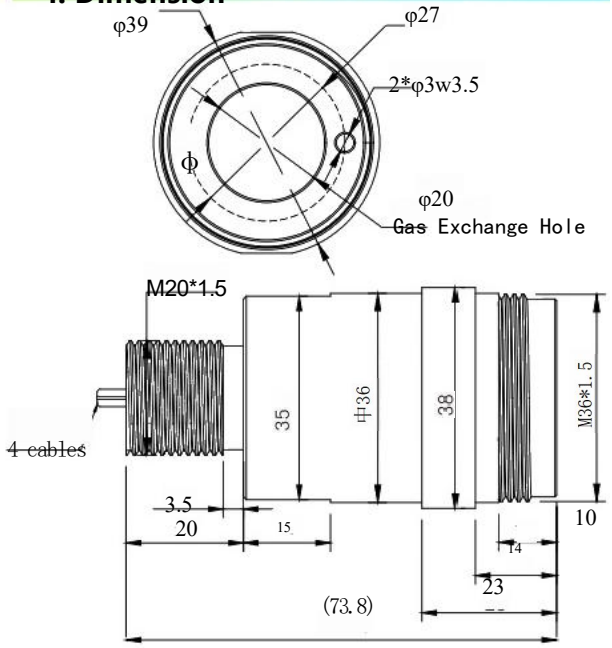
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3. Technical Parameter

Detection range	0	-	5	%Vol
(T90)	-	15	-	S
Resolution	0.01			%VOL
Communication Interface				
Standard TTL serial port	Baud Rate	115200		
	Stop Bit	1bit		
	Data Bit	8bit		
	Check Digit	-		
	Flow Control Bit	-		
Dimension				
Weight	<300g			
Dimension	ϕ39*L73.8mm			
Material	304 Stainless Steel			
IP	IP66			
Ex	Intrinsically safe			

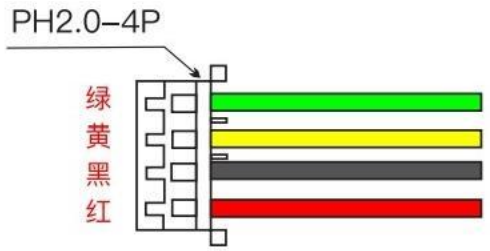
Parameter	Min.	Typical	Max.	Unit
Target Gas	Methane CH4			
Method	Diffusive			
Principle	TDLAS			
Electrical properties (room temperature)				
VCC	3.3	—	5.0	V
Current(mean)	—	80		mA
Current(Instant)	—	150	200	mA
General Parameters				
Working Temp.	-40	—	70	℃
Storage Temp.	-40	—	70	℃
Humidity (No condensation)	0	—	98	%RH
Working pressure	80	—	116	Kpa
Measurement error	Room Temperature and pressure		±0.15%	%VOL
	Full Load State		±0.35%	%VOL

4. Dimension



5. Interface

The lead length is about 15cm, and the lead connector is PH2.0-4P



NUMBER	Color	Definition
1	RED	VCC
2	BLACK	GND
3	YELLOW	RX
4	GREEN	TX

6. Communication Protocol

Output:Active fixed-length character ;

Data output interval：2s Data format ASCII string,29 bytes

Function	Byte	Byte position	Unit	Eg.(ASCII)
Concentration	1-7	7	%VOL	+000.00
Space	8	1	-	<SP>
Temp.	9-13	5	℃	+25.0
Space	14	1	-	<SP>
Pressure	15-21	7	KPa	+101.32
Space	22	1	-	<SP>
Status code	23-24	2	-	00
Space	25	1	-	<SP>
XOR	26-27	2	-	19
CR	28	1	-	<CR>
LF	29	1	-	<LF>

Data output format description:

+XXX.XX space nn.n space +ppp.pp space SS space HH<CR><LF>

+XXx.xx Concentration， unit：%VOL, range：+000.00~+999.99;

nn.n Temp.， unit：℃, range： -99.9~+99.9;

+ppp.pp air pressure，unit：KPa, range：+000.00~+999.99;

SS Status code ;

HH XOR checksum of the first 25 bytes，the check byte is output as 2 characters.;

<CR><LF>

Note:The temperature output by the sensor is the temperature compensation reference temperature, not the ambient temperature.

e.g. +000.00+22.3+101.670037

+000.00:CH4 concentration，round to two decimal places，Unit：%VOL;

+22.3:Temp.，unit：℃;

+101.67: air pressure，Unit： KPa;

00: Status code：

37: XOR

In default mode output, bytes 23 and 24 represent the status code in ASCII format, indicating the operational status of the probe. These two ASCII bytes are directly converted to 8-bit binary numbers according to hexadecimal conversion, with specific definitions as shown in the table below：

23				24			
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
retention	retention	retention	Temp. Control	Air Pressure	L-Light Intensity	H-Light Intensity	Absorption Peak Position
Default 0	Default 0	Default 0	0:Normal 1:Abnormal	0:Normal 1:Abnormal	0:Normal 1.Insufficient	0:Normal 1:Excessive	0:Normal 1.Abnormal

For example, if the sensor outputs the status code "11", which corresponds to "00010001" in binary, it indicates "Temperature Control Abnormality and Absorption Peak Position Abnormality.

7. Attention

- Do not drop the sensor from a height.
- Check the sensor's power supply and electrical connections before use to ensure they meet the requirements.
- Before use, ensure that the power cord is connected correctly and the voltage is normal.
- Do not disassemble or repair the sensor in explosive gas environments.
- Do not place the sensor near high temperatures or open flames.
- Do not perform soldering or similar operations on the sensor interface while it is powered.
- It is recommended to use a dedicated adapter for external wiring when performing independent tests.
- Do not attempt to disassemble or repair the sensor yourself.